

YI QING XU

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My research focuses on *translating* human objectives into signals for robotic optimization. I develop *compositional and hierarchical structures as intermediate representations* and design *reward learning algorithms* to better align robotic agents with human goals, especially when expert data is scarce or under-specified.

Keywords: Imitation Learning, Goal Specification for Robotics, IRL, Neuro-Symbolic Representation.

RESEARCH HIGHLIGHT

In my latest works, “**SetItUp**” ([IJRR](#)) and “**StackItUp**” ([CoRL](#)), I explore how robots can act on human goals conveyed through ambiguous but intuitive inputs — like language commands or freehand sketches. Both works share a neuro-symbolic architecture that maps these inputs into **abstract relation graphs**, then grounds them into feasible physical configurations via **compositional diffusion models**. This approach preserves task structure, supports generalization, and learns from surprisingly few demonstrations by reusing local relational priors.

I’m excited to extend this framework in two directions. First, toward **mixed-modality goal specifications** — combining coarse, abstract instructions with precise but partial demonstrations to infer symbolic task skeletons and modular reward functions that can be composed and optimized jointly. Second, toward **interactive goal specification interface**, where robots engage with users via language, gaze, and motion to resolve ambiguity through active dialogue and inference. Across both directions, the goal remains the same: to make goal specification more expressive, adaptable, and aligned with how humans actually communicate intent.

EDUCATION

National University of Singapore

Aug 02, 2020 - May 31, 2026

Ph.D in Computer Science, advised by [Prof. David Hsu](#)

GPA: 5.00

National University of Singapore

Aug 02, 2016 - Jun 30, 2020

B.A. in Computer Science

GPA: 4.81

B.A. in Applied Mathematics

GPA: 4.89

RESEARCH EXPERIENCE

Research Intern

Nov, 2025 – Present

Allen Institute for Artificial Intelligence (AI²), Robotics (Seattle), [Prof. Dieter Fox](#)

Collaborating with the robotics team to develop foundation models for the block world. My work focuses on task and goal specification—studying how human objectives can be represented, grounded, and translated into robot-executable actions.

Visiting Ph.D Student

Sep, 2023 - Feb, 2024

CSAIL MIT, [Prof. Leslie Kaelbling](#) and [Prof. Tomás Lozano-Pérez](#)

Developed “*Set It Up!*”, a neuro-symbolic system that interprets and optimizes under-specified instructions for tabletop arrangements using very few demonstrations. Published at RSS 2024.

SELECTED PUBLICATIONS / PREPRINTS

1. “Set It Up”: Functional Object Arrangement with Compositional Generative Models [\[Link\]](#)
Yiqing Xu, Jiayuan Mao, Linfeng Li, Yilun Du, Tomás Lozano-Pérez, Leslie Kaelbling, David Hsu
International Journal of Robotics Research (IJRR), 2025.
2. “Stack It Up”: 3D Stable Structure Generation from 2D Hand-drawn Sketch [\[Link\]](#)
Yiqing Xu, Linfeng Li, Cunjun Yu, David Hsu
Conference on Robot Learning (CoRL), 2025.
3. On the Effective Horizon for Inverse Reinforcement Learning [\[Link\]](#)
Yiqing Xu, Finale Doshi-Velez, David Hsu
International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2025.

4. “Set It Up!”: Functional Object Arrangement with Compositional Generative Models [\[Link\]](#)
Yiqing Xu, Jiayuan Mao, Yilun Du, Tomás Lozano-Pérez, Leslie Kaelbling, David Hsu
Robotics: Science and Systems (RSS), 2024
5. Differentiable Particles for General-Purpose Deformable Object Manipulation [\[Link\]](#)
Siwei Chen, Yiqing Xu, Cunjun Yu, Linfeng Li, David Hsu
In submission.
6. Grounding Common-sense Objective for Tabletop Object Rearrangement [\[Link\]](#)
Yiqing Xu, David Hsu
Robotics: Science and Systems (RSS), 2023, Lang2Rob Workshop
7. Learning Reward for Physical Skills using Large Language Models [\[Link\]](#)
Yuwei Zeng, Yiqing Xu
Conference on Robot Learning (CoRL), 2023, the 2nd Lang2Rob Workshop
8. Benchmarking Deformable Object Manipulation with Differentiable Physics [\[Link\]](#)
Siwei Chen, Yiqing Xu*, Cunjun Yu*, Linfeng Li, David Hsu*
International Conference on Learning Representations (ICLR): **Notable Top 5%**, 2023
9. Receding Horizon Inverse Reinforcement Learning [\[Link\]](#)
Yiqing Xu, Wei Gao, David Hsu
Advances in Neural Information Processing Systems (NeurIPS), 2022
10. Coach: Cooperative robot teaching [\[Link\]](#)
Cunjun Yu, Yiqing Xu, Linfeng Li, David Hsu
Conference on Robot Learning (CoRL), 2022

PROFESSIONAL SERVICE

Workshop Organizer

- Task Specification Workshop at RSS 2024 [\[Website\]](#) July 2024
- Second LEAP Workshop at CoRL 2024 [\[Website\]](#) November 2024

Reviewer: ICRA, RSS, AAMAS, AAAI, NeurIPS, ICLR, ICML

AWARDS

President’s Graduate Fellowship, National University of Singapore (2020).

NUS Research Achievement Award, National University of Singapore (2022, 2025).

INVITED TALKS & PRESENTATIONS

Invited Talk, Stanford Vision and Learning Lab, Stanford University (2025).

Invited Talk, Personal Robotics Lab, University of Washington (2025).

Invited Talk, Robotics Team, Allen Institute for Artificial Intelligence (AI2) (2025).

Oral Presentation, Conference on Robot Learning (CoRL) (2025).

MENTORSHIP

Connie Jiang (Summer Intern) — Now B.S., Massachusetts Institute of Technology (2023).

Aishik Pyne (M.S., NUS) — Now Founder, Stealth Startup (2024).

Wanchen Li (Ph.D., University of Montpellier) — Ongoing Research Mentorship (2025–present).